

DARES ELECTRONICS

BLDC Motor Control

Learning and Project Documentation Guide

PRINCIPLE -> SIMULATION -> HARDWARE -> TESTING -> CONTROL

A beginner-friendly reference for learning six-step ESC control, documenting engineering work, and progressing toward closed-loop control and field-oriented control.

CURRENT PRIORITY

BLDC principle -> three-phase inverter -> six-step commutation -> PWM -> simulation -> ODrive bring-up. Use ODrive to connect the theory to real closed-loop control before moving to a custom ESC and advanced FOC development.

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1. Purpose and scope

This guide gives new Motor Control members a shared starting point. It explains what to learn, what to simulate, what to document, and what evidence is required before the team moves from basic six-step commutation to closed-loop control and FOC.

By using this guide, a member should be able to:

- Explain how a BLDC motor produces torque and why electronic commutation is required.
- Explain the purpose of the three-phase inverter, gate driver, PWM, dead time, and sensing circuits.
- Build or review a basic six-step simulation and interpret the main voltage, current, speed, and torque results.
- Bring up an ODrive safely and use it to connect motor parameters, sensing, FOC, and closed-loop control to a real motor.
- Document schematic, firmware, simulation, and test work so another member can reproduce it.
- Recognise when the system is ready for speed feedback, current control, and FOC.

IMPORTANT BOUNDARY

This guide is an onboarding and documentation standard. It does not replace component datasheets, a design review, electrical safety procedures, or validation on the actual target motor and power system.

Recommended working method

Learn enough theory to understand the physical system, build a simple model, validate key ideas on an ODrive, compare the model with measured behaviour, and then deepen the control theory or develop custom hardware. The team should avoid treating theory, simulation, hardware, and testing as separate projects. Each step should create evidence for the next one.

2. Recommended learning path

The sequence below is designed for members with no previous motor-control experience. Complete each stage well enough to explain it in your own words before moving to the next stage.

1

BLDC motor operating principle

Understand where torque comes from.

A BLDC motor has a permanent-magnet rotor and three stator windings. Current in a winding creates a stator magnetic field. The controller changes which windings carry current so that the stator field advances around the motor. The rotor follows this field and produces mechanical rotation. Unlike a brushed DC motor, the commutation is performed electronically by the ESC.

Learning check

- Identify the rotor, stator, and three motor phases.
- Explain why current direction changes the direction of the stator magnetic field.
- Explain why a controller must know, or estimate, rotor position.

[YouTube: How does a Brushless DC Motor work?](#)

Animated introduction to the rotor, stator, three-phase windings, and electronic commutation.

2

Three-phase inverter and gate driver

Understand how DC battery power becomes controlled phase current.

The battery supplies DC power. A three-phase inverter uses three half-bridges, normally six MOSFETs, to connect each motor phase to the positive DC bus, the negative bus, or a floating state. The MCU generates logic-level commands; a gate driver supplies the voltage and current needed to switch the MOSFET gates. Dead time prevents the high-side and low-side devices in one half-bridge from conducting at the same time.

Learning check

- Explain high-side, low-side, half-bridge, and shoot-through.
- Explain why an MCU pin should not directly drive a power MOSFET gate.
- Identify the role of bootstrap circuitry, gate resistors, and dead time.

3

Six-step commutation

Understand the first practical control method.

In six-step commutation, two phases are driven and the third phase is normally floating. One driven phase carries current into the motor and the other carries current out. The controller advances through six electrical states so that the stator field moves around the motor. Reversing the order reverses the direction. The exact switching table depends on phase naming, Hall sequence, and desired rotation direction.

Learning check

- Describe the current path in all six states.
- Identify the high-side device, low-side device, and floating phase in each state.
- Explain why the sequence must be synchronised with rotor position.

[YouTube: Understanding Six-Step Commutation for BLDC Motors](#)

MathWorks introduction to the switching sequence and behaviour of a six-step BLDC drive.

4

PWM and power control

Understand how the controller changes average motor voltage.

Pulse-width modulation switches a MOSFET rapidly. Duty cycle is the fraction of each PWM period for which the command is active. Increasing duty cycle generally increases the average voltage available to the motor, but current and torque still depend on winding resistance, inductance, back-EMF, load, commutation timing, and the selected PWM strategy. PWM frequency is a compromise among current ripple, audible noise, switching loss, timer resolution, and control bandwidth.

Learning check

- Define PWM frequency, period, and duty cycle.
- Explain why duty cycle is not the same as guaranteed motor speed.

- Explain how dead time and switching losses affect real hardware.



Rotor position and back-EMF

Understand how commutation timing is determined.

Hall sensors provide coarse rotor-position sectors, while encoders provide finer mechanical position. A sensorless six-step controller can monitor the voltage of the floating phase and detect back-EMF zero crossings. Because back-EMF is small near zero speed, a sensorless ESC normally needs an open-loop starting sequence before it can transition to reliable closed-loop commutation.

Learning check

- Compare Hall, encoder, and sensorless position information.
- Explain why back-EMF increases with speed.
- Explain why low-speed sensorless starting is difficult.

3. Simulation before final hardware

Simulation is used to test understanding and reduce risk before the final power stage is energised. It is not a substitute for measurement. The first model should be simple enough that every block and parameter can be explained. Suitable tools include MATLAB/Simulink, PLECS, PSIM, or LTspice, depending on the required level of electrical and mechanical detail.

Minimum first simulation

- DC bus source with a documented voltage and current limit.
- Three-phase inverter or an equivalent switching model.
- BLDC motor model with documented electrical and mechanical parameters.
- Six-step commutation sequence and adjustable PWM duty cycle.
- Mechanical load or an explicitly stated no-load assumption.

Required outputs

- Phase voltages and phase currents.
- DC bus voltage and current.
- Motor speed and electromagnetic torque.
- Commutation state and PWM duty cycle.
- Start-up transient and steady-state behaviour.

What the report must explain

Do not upload plots without interpretation. State the objective, assumptions, input parameters, solver and sample time, expected behaviour, observed behaviour, and conclusion. If the model behaves incorrectly, record the failure and the evidence used to diagnose it. A failed simulation with a clear explanation is more useful than an unexplained successful screenshot.

SIMULATION EXIT CRITERION

A member can explain the current path for each commutation state and can account for the main voltage, current, speed, and torque waveforms. Changing duty cycle or load produces a physically reasonable response.

Official reference: MathWorks Six Step Commutation

Reference for six-step switching logic, Hall-state inputs, rotor-position inputs, and switching sequences.

4. Practical implementation with ODrive

ODrive provides a ready-made motor-control platform containing the power stage, gate drivers, current measurement, MCU, control firmware, communication interface, and rotor-feedback inputs. It allows the team to move from a simulated motor to a controlled physical motor without first designing a complete ESC. This makes it useful for learning motor calibration, current control, velocity control, position feedback, FOC, regenerative braking, and fault diagnosis.

IMPORTANT DISTINCTION

ODrive is not a six-step commutation teaching board. Its normal control approach is closed-loop current control and FOC. Six-step commutation should still be learned and simulated because it makes the inverter, phase-current paths, PWM, and rotor-position problem easier to understand. ODrive is the next practical step, not a replacement for those fundamentals.

How ODrive maps to the motor-control system

- DC input and bus capacitors provide energy to the inverter.
- The gate drivers and MOSFET bridges switch the three motor phases.
- Current shunts and amplifiers provide phase-current feedback for the inner current loop.
- The MCU executes calibration, FOC, protection, and torque, velocity, or position control.
- An encoder, Hall sensors, or a sensorless estimator provides rotor information, depending on the setup.
- USB, UART, CAN, or other interfaces allow configuration, commands, logging, and fault reporting.
- The brake resistor or another regeneration strategy manages energy returned to the DC bus during braking.

Record the exact hardware and firmware first

Before following any tutorial, record the ODrive model, voltage version, board revision, firmware version, host-tool version, and motor-feedback method. ODrive v3.x firmware 0.5.x and current ODrive products running 0.6.x do not use identical configuration names or procedures. Treat the video as a conceptual demonstration and use documentation that matches the exact board and firmware for actual commands.

Recommended bring-up workflow

- Inspect the board and wiring. Secure the motor, remove the propeller, and confirm the DC supply polarity and voltage range.
- Connect the three motor phases and the selected encoder or Hall feedback. Confirm pole pairs and feedback resolution from reliable sources.

- Set conservative bus-voltage, motor-current, calibration-current, velocity, and acceleration limits before enabling the axis.
- Plan for regenerated energy. Configure the correct brake resistor or regeneration limit for the specific board and power source.
- Run motor calibration so the controller can estimate phase resistance and inductance. Investigate errors rather than repeatedly increasing limits.
- Calibrate and verify the encoder or configure the intended sensorless method. Confirm that position and velocity change in the expected direction.
- Begin closed-loop operation at low torque and low speed. Test start, stop, direction, and loss-of-feedback behaviour before increasing the command.
- Log DC bus voltage, current, measured speed, target speed, controller mode, temperature, and error history for every meaningful test.

What members should learn from ODrive

- Why motor resistance and inductance are needed by a current controller.
- How pole pairs relate mechanical rotation to electrical angle.
- How encoder alignment and direction affect torque production.
- How torque, velocity, and position modes use nested control loops.
- Why current limits, velocity limits, bus-voltage limits, temperature limits, and fault states are part of the controller design.
- Why deceleration can return energy to the DC bus and cause over-voltage or regeneration faults.

Required ODrive evidence

- Board model, firmware version, power source, motor model, pole pairs, and feedback device.
- Saved configuration or a script that reproduces every changed parameter.
- Calibration results, including measured resistance and inductance where available.
- A low-speed closed-loop test with target and measured speed, bus voltage, current, and temperature.
- Complete error output and the reasoning used to diagnose each fault.
- A short conclusion explaining what the ODrive test proves and what it does not prove about the custom ESC.

YouTube: Unpacking and testing the new product ODrive

Practical introduction to connecting, configuring, and testing a BLDC motor with an ODrive controller.

Official ODrive reference: Motor Calibration Parameters

Explains calibration current and voltage, resistance and inductance measurement, and common calibration faults.

Official ODrive reference: Hardware Configuration

Covers power limits, regenerated energy, braking, temperature monitoring, encoders, and sensorless configuration.

ODRIVE EXIT CRITERION

The configuration is reproducible; calibration completes without an unexplained fault; the unloaded motor starts, runs, changes speed, and stops predictably at conservative limits; measured values are logged; and the member can relate the board behaviour to the inverter, sensing, and control concepts in this guide.

5. ESC hardware documentation

Hardware design should begin after the team understands the power path and has a basic model. The schematic must be divided into functional blocks, and every major component must have a documented selection reason. Copying a reference design is acceptable only when the team verifies that its voltage, current, thermal, and control assumptions match the project.

MCU and timing

PWM channels, complementary outputs, dead-time support, ADC triggering, interrupt timing, communication, and available processing margin.

Gate driver

High-side and low-side drive capability, bootstrap limits, UVLO, fault reporting, propagation delay, peak gate current, and supported switching frequency.

MOSFET power stage

Drain-source voltage margin, current and pulse ratings, $R_{DS(on)}$, gate charge, switching loss, thermal resistance, package, and PCB current paths.

Current sensing

Shunt location, resistance, power rating, amplifier range and bandwidth, common-mode limits, ADC range, filtering, and over-current protection.

Voltage sensing

Battery and DC bus dividers, phase-voltage or back-EMF scaling, filtering, transient protection, and ADC input limits.

Power supplies

Logic rails, gate-driver supply, regulator input range, start-up behaviour, decoupling, and power sequencing.

Protection and thermal design

Reverse polarity, input transients, shoot-through prevention, over-current, under-voltage, over-voltage, over-temperature, fusing, and heat removal.

Interfaces

Rotor sensor inputs, throttle or command input, telemetry, programming, debug, and fault/status outputs.

Component selection record

For each critical component, record the selected part, required rating, calculated margin, relevant datasheet pages, alternatives considered, reason for selection, risks, and validation plan. The absolute maximum rating is not a normal operating target; design against recommended operating conditions and expected transients.

6. Firmware documentation

Firmware documentation must describe the control timing and the relationship between software commands and physical switching. A reader should be able to trace a target command through the commutation logic, PWM generation, ADC sampling, feedback calculation, protection checks, and fault response.

Minimum topics

- Timer, PWM, complementary output, dead-time, and update-event configuration.
- Six-step state machine, direction control, and commutation timing.
- Hall, encoder, or back-EMF input processing and validation.
- ADC trigger timing, scaling, offset calibration, and filtering.
- RPM calculation and timeout behaviour.
- Start-up, run, stop, and fault states.
- Over-current, over-voltage, under-voltage, and over-temperature responses.
- Communication protocol, units, limits, and error handling.

Function-level documentation

Important functions should state their purpose, inputs, outputs, units, valid range, call rate, side effects, and safety assumptions. Constants such as pole pairs, current-sensor gain, ADC reference, timer clock, and PWM frequency must have one authoritative definition. Changes to these constants should be reviewed because they can alter physical limits.

FIRMWARE EXIT CRITERION

The motor can be commanded to start and stop predictably; the commutation sequence and feedback units are verified; every defined fault forces a documented safe state; and the firmware version is recorded with each test.

7. Bench testing and safety

The first practical objective is deliberately narrow: rotate the motor reliably using six-step commutation and PWM. Test one assumption at a time and capture the evidence. Start with logic-level and gate-signal checks before connecting the motor to the complete power stage.

Minimum safety controls

- Remove the propeller for controller development unless an approved guarded thrust test specifically requires it.
- Use a current-limited supply where practical, plus an appropriate fuse and accessible emergency disconnect.
- Secure the motor and loose wiring before energising the system.
- Verify PWM polarity, complementary outputs, dead time, and gate waveforms before applying full bus voltage.

- Use differential or isolated measurement methods where required; do not attach an earth-referenced oscilloscope ground to an unsafe switching node.
- Begin at low voltage and conservative current limits, then increase only after reviewing measurements.
- Monitor MOSFET, gate driver, motor, connector, and cable temperature.
- Stop immediately for unexpected current, noise, heating, loss of commutation, or unstable control.

Test record

- Test ID, date, operator, reviewer, and objective.
- Schematic/PCB revision, firmware commit, configuration, and instrument list.
- Power supply voltage and current limit, motor, sensor, load, and propeller status.
- Procedure, expected result, measured result, plots or oscilloscope captures, and pass/fail decision.
- Unexpected behaviour, likely cause, follow-up action, and link to the issue or design note.

FIRST HARDWARE MILESTONE

The motor starts, continues rotating, and stops on command. Current remains within the planned limit, temperatures remain acceptable, gate signals show no shoot-through, and the result is reproducible.

8. Closed-loop speed and current control

Speed feedback

Open-loop control applies a command but does not guarantee the requested speed. Closed-loop speed control compares target RPM with measured RPM and adjusts the motor command to reduce the error. The documentation must state how RPM is measured, the update rate, filtering, units, valid range, and behaviour when feedback is lost.

PID understanding

Proportional action reacts to present error. Integral action accumulates error and can remove steady-state offset, but it needs output limiting and anti-windup. Derivative action reacts to the rate of change and is sensitive to measurement noise. A PI controller is often sufficient for speed control; the correct choice depends on the plant, sample rate, noise, and performance requirement. Gains must be recorded with the test conditions and not treated as universal constants.

Current control

Motor torque is strongly related to current, so a current loop provides more direct torque control and protects the power stage. In a cascaded controller, the speed loop normally generates a current or torque request, while the faster inner current loop controls the electrical response. The selected current-sensing topology and PWM/ADC timing determine which measurements are valid.

Required control evidence

- Target, measured value, error, controller output, saturation limits, and sample time.
- Step response under at least two loads or operating points.
- Rise time, overshoot, settling behaviour, steady-state error, and oscillation/noise observations.

- Controller gains, tuning method, anti-windup method, filters, and safety limits.

9. Introduction to field-oriented control

FOC should begin only after phase-current measurement, rotor angle, PWM timing, and basic feedback control are understood. FOC transforms measured three-phase currents into a rotating reference frame aligned with the rotor. This produces d-axis and q-axis current components that can be controlled using fast PI loops. In simplified terms, q-axis current is mainly associated with torque, while d-axis current is associated with flux. The commanded voltages are transformed back and applied through a modulation method such as SVPWM.

Concepts to learn

- Mechanical angle, electrical angle, and pole pairs.
- Phase-current reconstruction and current-sampling windows.
- Clarke and Park transforms, plus their inverse transforms.
- d-axis and q-axis current references and PI current controllers.
- Voltage limiting, decoupling, and PI anti-windup.
- SVPWM, encoder alignment, and sensorless observers.

YouTube: Basics of Field-Oriented Control

STMicroelectronics introduction to BLDC/PMSM control, FOC, current sensing, and sensor/sensorless topologies.

YouTube: Field-Oriented Control on STM32 From Scratch

Practical implementation reference covering encoder feedback, current measurement, transforms, and SVPWM.

Official course: STM32 Motor Control Part 5

ST's structured FOC course, including theory, STM32 implementation, current sensing, observers, and practical tools.

FOC ENTRY CRITERION

The team has trustworthy current measurements, a verified electrical angle, deterministic PWM and ADC timing, safe current limits, and a working test method. FOC should not be used to hide unverified hardware behaviour.

10. Repository structure

The repository should make the current design state easy to find. The main README is an entry point; detailed technical work belongs in focused Markdown documents and version-controlled project files. Suggested paths are:

```
README.md - project overview, status, and quick start
docs/motor-basics/ - learning notes and reviewed explanations
docs/control-theory/ - speed, current, FOC, and tuning notes
docs/datasheets/ - links or approved copies of key datasheets
docs/design-notes/ - engineering decisions and trade-offs
simulation/ - model files, parameters, and reports
hardware/schematic/ - schematic source, exports, and review notes
hardware/pcb/ - layout source, manufacturing files, and reviews
firmware/ - source code, configuration, and build instructions
testing/odrive/ - board configuration, scripts, logs, and fault records
testing/ - test plans, raw data, logs, and reports
results/ - approved summary plots and milestone evidence
```

File and commit expectations

- Use descriptive filenames and include units or dates where they prevent ambiguity.
- Do not upload an unexplained screenshot, model, binary, or data file.
- Link research sources and identify the exact datasheet revision used.
- Record failures as well as successes; do not remove evidence simply because a test failed.
- Keep generated exports separate from editable source files.
- Use issues or design notes for unresolved questions and link the relevant commit or test report.
- Request review for changes that affect switching, sensing, current limits, protection, or test safety.

11. Required document templates

Simulation note

- Objective and question being tested.
- Model scope, assumptions, parameter sources, solver, and sample time.
- Inputs and expected behaviour.
- Results with units and explanation.
- Limitations, problems, conclusion, and next action.

Design decision record

- Decision and date.
- Requirement or problem being addressed.
- Alternatives considered and evidence reviewed.
- Selected option, calculations, margins, benefits, and disadvantages.
- Risks, validation plan, owner, and review status.

Test report

- Test ID, objective, operator, reviewer, date, and location.
- Hardware revision, firmware commit, configuration, motor, load, supply, and instruments.
- Safety controls and pre-test checks.

- Procedure, expected result, measured data, and uncertainty where relevant.
- Pass/fail decision, fault observations, conclusion, and linked follow-up issue.

Datasheet review

Before using a component, review at least the absolute maximum ratings, recommended operating conditions, pin descriptions, electrical characteristics, timing requirements, thermal information, and typical application circuit. Record which limits are design requirements and which must be validated during testing.

12. First tasks for new members

The following tasks turn passive learning into reviewable engineering output. New members should work with an experienced member, but they should still form and test their own explanation.

Task 1 - Principle note

Write a short explanation of how a BLDC motor, inverter, six-step sequence, and PWM work together. Include the current path for each commutation state in words or a switching table.

Task 2 - Simulation

Create or reproduce a basic six-step simulation. Change duty cycle and load, then explain the effect on current, speed, and torque.

Task 3 - ODrive bring-up

Record the exact board and firmware, create a safe configuration, complete calibration, run a conservative closed-loop speed test, and document all measured values and faults.

Task 4 - Hardware block review

Choose one block - gate driver, MOSFET stage, current sensing, voltage sensing, power supply, or protection - and document its requirements and one candidate implementation.

Task 5 - GitHub submission

Upload the editable source, a readable result, the ODrive configuration or script, and a Markdown explanation. Link every source and state what remains uncertain.

Task 6 - Short presentation

Present what was learned, one mistake or unexpected result, how it was investigated, and the next test that would reduce uncertainty.

REVIEW STANDARD

The goal is not to appear correct on the first attempt. The goal is to make assumptions visible, test them, record the evidence, and improve the next design.

13. Milestones and definition of done

Milestone 1 - Fundamentals

A member can explain the motor, inverter, PWM, six-step current paths, and position feedback without copying a diagram or script.

Milestone 2 - Simulation

The model runs, inputs and parameters are documented, waveforms are physically reasonable, and results are explained.

Milestone 3 - ODrive reference-platform bring-up

The ODrive configuration is reproducible; calibration is understood; the motor runs under conservative closed-loop limits; data and faults are documented.

Milestone 4 - Open-loop or commutated custom bench run

The motor starts, runs, and stops safely; current and temperature stay within limits; gate signals and commutation are verified; the test is reproducible.

Milestone 5 - Closed-loop speed

RPM feedback is validated; the controller tracks commands over the agreed operating range; gains, limits, and response measurements are documented.

Milestone 6 - Current control and FOC readiness

Current sensing and electrical angle are verified; PWM/ADC timing is deterministic; protection is tested; the team has an agreed FOC validation plan.

Project-level definition of done

- Editable design sources and build instructions are committed.
- Requirements, calculations, assumptions, and safety limits are documented.
- The relevant simulation and test evidence is stored with units and version information.
- Known limitations and unresolved risks are explicit.
- Another team member can reproduce the result without relying on the original author.
- A reviewer has approved the milestone against its stated exit criteria.

14. Learning resources

Use the resources below as starting points, then confirm implementation details against the relevant motor, MCU, gate-driver, MOSFET, sensor, and regulator datasheets. Video explanations are useful for intuition but are not the final authority for component limits or safety decisions.

[YouTube - How does a Brushless DC Motor work?](#)

Beginner animation for motor construction and electronic commutation.

YouTube - Understanding Six-Step Commutation for BLDC Motors

Six-step switching and simulated motor behaviour from MathWorks.

MathWorks - Six Step Commutation reference

Official switching-sequence reference for Hall or position inputs.

YouTube - Unpacking and testing the new product ODrive

Practical ODrive connection, configuration, calibration, and motor test introduction.

YouTube - Basics of Field-Oriented Control

FOC introduction from STMicroelectronics.

STMicroelectronics - Motor Control Part 5

Structured course covering FOC theory, STM32 implementation, sensing, and observers.

YouTube - Field-Oriented Control on STM32 From Scratch

Practical implementation reference for current sensing, encoder feedback, transforms, and SVPWM.

Useful search terms

- BLDC motor working principle
- three-phase inverter and half-bridge explained
- MOSFET gate driver dead time and shoot-through
- BLDC six-step commutation
- BLDC PWM strategies
- back-EMF zero-crossing sensorless BLDC
- ODrive motor calibration, encoder setup, regeneration, and brake resistor
- BLDC speed PI control
- Clarke Park transform motor control
- FOC current sensing and SVPWM

Start this week

- Watch the BLDC principle, six-step commutation, and ODrive bring-up videos.
- Write a one-page explanation in your own words.
- Build or review a basic six-step simulation and change one parameter.
- Create a version-specific ODrive safety checklist, then plan one conservative closed-loop test.
- Upload the editable source and evidence, then present one unresolved question and the next test.